

VINAY RAJU SANNIBOINA

AUTOMOTIVE ENGINEER

MSc Automotive Engineering (Electric Vehicles) student, graduating 05/2026, with a background in mechanical design, vehicle dynamics, and simulation-driven analysis across CAD, FEA, and MATLAB/Simulink. As part of Formula Student Bedfordshire Racing, took Design from concept to manufacture and track testing, working to weight, packaging, and cost targets against a fixed competition deadline. Looking to develop broadly across vehicle engineering on a graduate programme, with a particular interest in Manufacturing Systems Simulation & Quality Assurance.

Education

University of Bedfordshire, Luton, UK.

May 2025 – May 2026

MSc Automotive Engineering for Electric Vehicles

Student: Gained knowledge.

- Thermodynamics, Heat Transfer & Fluid Dynamics
- Computational Fluid Dynamics (CFD)
- Vehicle Dynamics & Performance
- Automotive Electronics & Control Systems
- Engineering Analysis & Numerical Modelling

Dissertation

Aerodynamic Performance Analysis and Optimisation of a Multi-Element Formula Student Front Wing Using Computational Fluid Dynamics (CFD)

- Conducted aerodynamic analysis and optimisation of a multi-element Formula Student front wing using **ANSYS Fluent** and **Space Claim**.
- Designed and refined front wing geometries in **CAD**, evaluating the influence of aerodynamic parameters on downforce, drag, and overall vehicle performance.
- Performed **mesh independence studies**, applied appropriate turbulence models, and analysed airflow characteristics including pressure distribution and velocity contours.
- Achieved approximately **45% reduction in aerodynamic drag** while improving aerodynamic efficiency through iterative design optimisation. Interpreted CFD simulation results to support engineering decisions and enhance design performance, demonstrating strong analytical, problem-solving, and numerical modelling skills.

Produced a comprehensive technical dissertation, communicating methodology, simulation results, and engineering recommendations in accordance with academic and industry standards.

Technical Skills Applied

ANSYS Fluent | Space Claim | SolidWorks | CFD | Aerodynamics | Vehicle Dynamics | Mesh

- vinayraj.3611@gmail.com
- www.linkedin.com/in/vinay-raju-sanniboina-474b79220

- +44 790042005

Education: MSc Automotive Engineering for Electric Vehicles (2:1)

University of Bedfordshire, Luton,
United Kingdom | **2025–2026**

Dissertation: Aerodynamic Performance Analysis and Optimisation of a Multi-Element Formula Student Front Wing Using Computational Fluid Dynamics (CFD)

BEng Electrical & Electronics Engineering (Specialisation in Electric Vehicles) – First Class Honours

Bharath Institute of Higher
Education and Research, India |
2020–2024

MEMBERSHIPS

- IMechE

Full UK Driving License

LANGUAGES

- English – Full Professional
- Telugu – Mother language

COMPUTER SKILLS

- SOLIDWORKS
- Simulation: ANSYS, CFD
- EV Systems: Battery systems, power electronics, motor control
- Tools: Microsoft Excel, Word

STUDENT:

Relevant Areas of Study

- Electrical Circuits & Power Systems
- Electrical Machines & Power Electronics
- Control Systems & Embedded Systems
- Automotive Electronics
- Electric Vehicle Technologies & Battery Systems
- Engineering Analysis & Problem Solving

Formula Student

Team Bedfordshire Racing, LUTON, UK

Position: Formula Student Team Member

Sep 2025 – June2026

Class II Vehicle project

Active member of the Formula Student team contributing to the design, development, and quality assurance of race car components. Involved in 3D CAD modelling, aerodynamic evaluation, and performance-driven design improvements under strict competition regulations. Collaborated with manufacturing and assembly teams to ensure components met performance, reliability, and compliance standards while working within competitive deadlines.

- Designed and modelled automotive components using SOLIDWORKS
- Contributed to aerodynamic evaluation and performance optimisation
- Collaborated with manufacturing teams to ensure design feasibility
- Supported quality assurance and compliance with competition standards